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Unit – 4 : Session – 11 : SLO – 2 | Odd Man Out — Pipeline Hazards

■ I. Odd Man Out

1. Group RAW, WAR, WAW, RAR → RAR — two reads never conflict; the other three are true data hazards.
2. Group Data forwarding, Branch prediction, Operand caching, Disk defragmentation → Disk defragmentation — a storage maintenance task, not a hazard technique.
3. Group Structural, Control, Data hazard, Power surge → Power surge — an electrical fault, not a pipeline hazard.
4. Group Bubble, Flush, Stall, Acceleration → Pipeline acceleration — the others are disruption terms.
5. Group ALU, FPU, Cache memory, Branch target buffer → Cache memory — a storage structure; the others are processor execution/prediction hardware units.
6. Group Bypassing, Speculative execution, Out-of-order execution, Sequential processing → Sequential processing — no overlap/parallelism.
7. Group Fetch, Decode, Execute, Compile → Compile — a software step, not a pipeline stage.
8. Group CPI, IPC, MIPS, USB → USB — an I/O interface; the others are performance metrics.
9. Group NOP insertion, Register renaming, Reorder buffer, Motherboard → Motherboard — a physical board; the others are hazard-handling mechanisms.
10. Group RISC pipeline, CISC pipeline, Printer driver → Printer driver — peripheral software.